Scott J. Gilbert

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Aspiring computer engineer with strong foundations in hardware design, embedded systems, and full-stack development. Skilled in C/C++, Python, systems engineering, low-level programming, and data processing, with hands-on experience in real-time systems, machine learning, and DevOps.

Education

University of Illinois Urbana-Champaign **May 2029 | GPA:** 4.00/4.00 (Dean's List) **| Chancellor's Scholar**

Bachelor of Science in Computer Engineering — The Grainger College of Engineering

Minor in Business — *Gies College of Business*

Activities and Societies: Campus Honors Program, Illini Solar Car, Project: Code, Illini Redstone Computing, Engineers without Borders, Marching Illini

Relevant Coursework: Differential Equations, Computational Linear Algebra, Computers & Programming, Quantum Physics, Thermodynamics, Discrete Mathematics, Data Structures, Analog Signal Processing, Fundamentals of Accounting

Professional Experience

Computer Administrator **Oct 2025 - Present**

University of Illinois Campus Honors Program (CHP) Urbana, Illinois

- Support core CHP operations by maintaining hardware infrastructure, external storage integrations, and web services across various providers.
- Coordinate with 15+ CHP staff, IT staff, and other administrators to modernize computing resources and increase accessibility/utility to students.
- Synchronize data between university admissions and CHP databases, reducing entry time for critical record processing by up to two weeks.

Full-Stack/AI Intern **June 2025 - Aug 2025**

Team2Go Inc. Yongin-si, Gyeonggi-do, South Korea (Remote)

- Collaborated with my intern team to deploy a full stack AI application publishing platform supplemented by Retrieval-Augmented Generation, a custom chat interface, rigid boundary/instruction declaration, virtualization in Docker, and full localization in English and Korean.
- Explored and deployed prompt-engineering techniques to reduce LLM hallucination and improve accuracy without sacrificing ease of deployment.
- Reported regular progress to senior executives and shipped key deliverables two weeks ahead of schedule.

Leadership/Involvement

Array/Maximum Power Point Tracking (MPPT) Lead & Webmaster **May 2026 - Present**

Illini Solar Car Champaign-Urbana, Illinois

- Co-led design of ISC's most efficient array (1.5 kW) by configuring MPPT firmware, modeling sun exposure in OpenGL, and managing ~$45K budgets.
- Modernized firmware to support new Arm Cortex MCUs, I2C expander removal, and CAN messaging and coordinated development, testing, and/or integration (enclosures, cooling, wiring) of key systems like MPPTs, wheel, and dash that powered the car 650+ miles without failure.
- Assist development and deployment of strategy/telemetry platforms, including cloud-based message brokers, databases, and web applications.
- Serve on the executive board, maintaining mission-critical software (MediaWiki knowledgebase, institutional/outreach website, communication platforms, etc.) across hosting providers. Manage virtual machines, DNS, etc. and address critical failures and other service requests on call.

Backend Team Lead **Sep 2025 - Dec 2025**

Project: Code UIUC (Agri-Sense Project) Champaign-Urbana, Illinois

- Managed team of ten developers focused on deploying data handling, APIs, and core logic for a data-driven, all-in-one crop monitoring platform.
- Oversaw integration of UART, OneWire, and I2C data buses on IoT boards to communicate sensor readings and other metadata to central backend.
- Maintained production branches, versioning and CI policies, and shared progress/API configuration with project and hardware/frontend leads.

Eagle Scout **Mar 2018 - Dec 2024**

Scouting America Hoffman Estates, Illinois

- Led restoration of 1500+ square feet of nature trail for my Eagle Scout Project, coordinating key elements from materials sponsorship and design to volunteer safety training and logistics. 20+ volunteers contributed 100+ working hours with zero safety incidents.
- Earned 27 merit badges, 10+ other certifications/awards, and held 5 leadership positions, including troop guide and assistant senior patrol leader.
- Practiced and continue to practice Scout Oath and the 12 points of the Scout Law in service to the community and to the nation.

Projects

Illini Redstone Computing (Engineering Club) Infrastructure **Jan 2026 - Present**

- Designed IRC's computing backbone, deploying three container orchestrations and various other services across four different hosting providers.
- Integrated gaming servers, WebDAV, DNS, web-based administration, datastores, messaging services, staff/admin authentication, API gateways, and secure connection tunneling into the full IRC stack, keeping services modular to support future upgrades/expansions.

Personal Content System **May 2025 - Present**

- Deployed microservices-based stack incorporating portfolio, blog, CMS, account management, workflow automation, CI/CD, and backend services.
- Increased CTR/search visibility via custom DNS/TLS configuration, >95 scores on core web vitals, and fine-grained SEO control.
- Wrote e2e test cases validating database seeding, API schema/guardrails, and deployment on edge/serverless environments, among other areas.

Skills

- **Programming:** C, C++, Python, SQL, Java, Assembly, JavaScript, TypeScript, PHP, Wolfram
- **Cloud/Data:** Docker, Podman, Vercel, MySQL, SQLite, MariaDB, PostgreSQL, pgvector, S3, UploadThing, Backblaze, CSV
- **Numerical Analysis:** Jupyter Notebook, NumPy, SciPy, Matplotlib, Mathematica, Excel, Google Sheets
- **Embedded Systems/Microcontrollers:** NXP, Arm Cortex, Mbed OS, AMD Vivado**,** Arduino, ELEGOO, CAN, UART, I2C, OneWire, KiCad,
- **DevOps, Tools, & Automation:** Vercel, Claude Code, Git, GitHub, GitHub Copilot, GitHub Actions, n8n, Make, MCUXpresso, JetBrains
- **Technical Expertise:** Embedded Systems, Firmware Design, Backend Development, PCB Design, AI/ML Application, Virtualization/Containerization

Awards: National Merit Scholarship Recipient, AP Scholar with Distinction, Gold President's Volunteer Service Award, Seal of Biliteracy (Mandarin)